DEDUCTION · LESSON 3

Truth Tables

Name: ____________________ Date: ____________

Patience rule: Build tables column by column, left to right. A final column copied from memory without sub-columns will not count. Circle or mark each row as you finish it so you can see you did not skip one.

Deduction problems — Truth tables

D3.1 DEDUCTION

Build a complete 8-row truth table for:

$((p \vee q) \wedge \neg r) \vee (p \wedge q \wedge r)$

Required columns, in order: p, q, r, $p \vee q$, $\neg r$, $(p \vee q) \wedge \neg r$, $p \wedge q$, $p \wedge q \wedge r$, final $\vee$.

1. Fill every cell.
2. List the row numbers (1–8) where the final column is T.
3. Describe in plain English a single real-world situation that matches one true row and one false row (different rows).

FULL TABLE + ROW LIST + TWO ENGLISH STORIES

D3.2 DEDUCTION

Determine whether each formula is a **tautology**, **contradiction**, or **contingency**. You must build a full table for each — no shortcuts from memory.

a. $(p \wedge (q \vee \neg q)) \vee (\neg p \wedge q)$

b. $(p \vee q) \wedge (\neg p \vee \neg q) \wedge (p \vee \neg q)$

c. $\neg(p \vee (q \wedge \neg p)) \vee (p \wedge q)$

For (b), the table uses only p and q (4 rows). For (a) and (c), use 4 rows as well.

THREE TABLES + LABEL EACH TAUTOLOGY/CONTRADICTION/CONTINGENCY

D3.3 DEDUCTION

Prove or disprove logical equivalence by full tables:

Left: $(p \to q) \wedge (p \to \neg q)$ Right: $\neg p$

(Recall: $p \to q$ is $\neg p \vee q$ when building columns.)

1. Build one 4-row table with separate columns for each side. Include columns for both conditionals on the left.
2. State equivalent or not.
3. Explain in English what the left side means (“If p then... and if p then...”) and why that matches or clashes with $\neg p$.

TABLE + EQUIVALENCE VERDICT + ENGLISH EXPLANATION

D3.4 DEDUCTION

Design challenge — read carefully:

Find a compound using exactly **four** connective symbols (each ¬ counts as one; each ∧, ∨ counts as one) over p, q, r such that the final column is T in **exactly three** of the eight rows.

1. Write your candidate formula.
2. Verify with a full 8-row table (no gaps).
3. Explain why the three true rows are the only ones — walk row by row.
4. Verify you used exactly four connective symbols (underline each).

FORMULA + VERIFICATION + ROW-BY-ROW WALK + CONNECTIVE COUNT